

# HW 04 Solutions

1/

1. (a) By inspection,

$$r(x) = 2p_0 + 3p_1 - p_2$$

$$\text{and thus } [r(x)]_S = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$$

(b) We seek coefficients  $\alpha_0, \alpha_1, \alpha_2$  such that

$$r(x) = \alpha_0 q_0 + \alpha_1 q_1 + \alpha_2 q_2,$$

hence

$$\begin{aligned} 2 + 3x - x^2 &= \alpha_0 + \alpha_1(1-x) + \alpha_2(x+x^2) \\ &= (\alpha_0 + \alpha_1) + (\alpha_2 - \alpha_1)x + \alpha_2 x^2. \end{aligned}$$

Therefore

$$\alpha_0 + \alpha_1 = 2$$

$$\alpha_2 - \alpha_1 = 3$$

$$\alpha_2 = -1$$

which yields

$$\alpha_0 = 6$$

$$\alpha_1 = -4$$

$$\alpha_2 = -1$$

In other words

$$[r(n)]_Q = \begin{bmatrix} 6 \\ -4 \\ 1 \end{bmatrix}$$

~~(2)~~

(2/5)

(c) We define  $\bar{P} = P^{-1}$ , and compute

$\bar{P} = [\bar{P}_1, \bar{P}_2, \bar{P}_3]$ , where  ~~$\bar{P}$~~

$$\bar{P}_1 = [q_0]_S = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\bar{P}_2 = [q_1]_S = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$\bar{P}_3 = [q_2]_S = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{Hence, } P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

2. By inspection, the e-values are  $\lambda_1 = 1$ ,  $\lambda_2 = 2$ ,  $\lambda_3 = 3$ .

$$(A_3 - \lambda_1 I) = \begin{bmatrix} 0 & 4 & 10 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$0 = (A_3 - \lambda_1 I)v^1 \Rightarrow v^1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$(A_3 - \lambda_2 I) = \begin{bmatrix} -1 & 4 & 10 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$0 = (A_3 - \lambda_2 I)v^2 \Rightarrow v^2 = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$$

$$(A_3 - \lambda_3 I) = \begin{bmatrix} -2 & 4 & 10 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$0 = (A_3 - \lambda_3 I)v^3 \Rightarrow v^3 = \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$$

The e-vectors are NOT unique.

Any non-zero multiples are also e-vectors.

Because the e-values are distinct, the e-vectors should be linearly independent.

$$0 = \alpha_1 v^1 + \alpha_2 v^2 + \alpha_3 v^3 = \begin{bmatrix} v^1 & v^2 & v^3 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

$$V = [v^1 | v^2 | v^3] = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and  $\det(V) = 1 \neq 0$ . Hence, only

solution is  $\alpha_1 = \alpha_2 = \alpha_3 = 0$ , and

indeed, the set  $\{v^1, v^2, v^3\}$  is linearly independent.

Insert this into the solution

5/

Prob2.txt

See folder

```
\begin{verbatim}
>> A3=[1 4 10; 0 2 0; 0 0 3];
>> [V,D]=eig(A3)
```

V =

```
    1.0000    0.9701    0.9806
           0    0.2425           0
           0           0    0.1961
```

D =

```
    1    0    0
    0    2    0
    0    0    3
```

```
\end{verbatim}
```

MATLAB normalizes the e-vectors so that the sum of the squares of the components equals 1.0. Here is how to check our answers

```
\begin{verbatim}
```

```
>> (A3-eye(3))*[1 0 0]'
```

ans =

```
    0
    0
    0
```

```
>> (A3-2*eye(3))*[4 1 0]'
```

ans =

```
    0
    0
    0
```

```
>> (A3-3*eye(3))*[5 0 1]'
```

ans =

```
    0
    0
    0
```

```
\end{verbatim}
```

6/

3. By inspection,  $\lambda_1 = 3, \lambda_2 = 3, \lambda_3 = 2$   
and we have repeated e-values.

$$(A_4 - \lambda_1 I) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

$$0 = (A_4 - \lambda_1 I)v' \Rightarrow v' = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Of ~~z~~ course, the same is true for

$\lambda_2$ , and thus  $v^2$  is not independent  
of  $v^1$ .

∴ No basis of e-vectors is possible.

4. We use two facts:

$$(i) \quad P^{-1} \cdot P = I$$

$$(ii) \quad \det(M_1 M_2) = \det(M_1) \det(M_2)$$

for compatible matrices  $M_i$ .

$$\begin{aligned} \det(\lambda I - B) &= \det(\lambda I - P^{-1} A P) \\ &= \det(\lambda P^{-1} P - P^{-1} A P) \\ &= \det(P^{-1} [\lambda I - A] P) \\ &= \det(P^{-1}) \det(\lambda I - A) \det(P) \end{aligned}$$

$$\text{But } 1 = \det(I) = \det(P^{-1} P) = \det(P^{-1}) \det(P),$$

and thus

$$\det(\lambda I - P^{-1} A P) = \det(\lambda I - A).$$



5. We translate  $Av^i = \lambda_i v^i$   
into the matrix formula

$$AP = P\Lambda$$

where  $P = [v^1 \dots v^n]$  and  $\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$ .

$P$  is invertible if, and only if, its  
columns are linearly independent,  
which we know is the case when  
the  $e$ -values are distinct.

$$\therefore P^{-1}AP = \Lambda$$

is the general result, and the specific  
result for  $A_3$  is

$$\begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 4 & 10 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Remark You did not have to  
provide the general result!

6.  $A = [A_1 | A_2 | A_3 | A_4]$  where

$$A_i = [L(u_i)]_{\{u_j\}}$$

$$\{u_j\} = \{E^{11}, E^{12}, E^{21}, E^{22}\}.$$

We compute

$$L(E^{11}) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = 1 \cdot E^{11} + 0 E^{12} + 0 E^{21} + 0 E^{22}$$

$$L(E^{12}) = \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix} = 0 E^{11} + \frac{1}{2} E^{12} + \frac{1}{2} E^{21} + 0 E^{22}$$

$$L(E^{21}) = L(E^{12})$$

$$L(E^{22}) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = 0 E^{11} + 0 E^{12} + 0 E^{21} + 1 E^{22}$$

so

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

7.

(a) We use the basis

$$\left\{ e^1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e^n = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix} \right\}$$

$$\hat{A}_i := [L(e^i)]_{\{e^j\}}, \quad \hat{A} = [\hat{A}_1 \dots \hat{A}_n]$$

$L(e^i) = Ae^i =$  the  $i$ -th column of  $A$ .

Hence  $\boxed{\hat{A} = A}$ .

(b) Let  $\{\lambda_1, \dots, \lambda_n\}$  be the  $e$ -values, which are assumed to be distinct, and let  $\{v^1, \dots, v^n\}$  be the  $e$ -vectors, which are consequently linearly independent and form a basis

12

for  $(\mathbb{C}^n, \mathbb{C})$ . We define

$$\hat{A}_i = [L(v^i)]_{\{v^k\}}.$$

$$L(v^i) = Av^i = \lambda_i v^i, \text{ and thus}$$

$$\hat{A}_i = \lambda_i e^i$$

and therefore

$$\hat{A} = \hat{L} = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

8. (a) We write  $\hat{f}(t_i) = \alpha_0 + \alpha_1(t_i) + \dots + \alpha_5(t_i)^5$   
and form the matrices

$$A = \begin{bmatrix} 1 & t_1 & (t_1)^2 & \dots & (t_1)^5 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & t_n & (t_n)^2 & & (t_n)^5 \end{bmatrix}$$

$$b = \begin{bmatrix} f(t_1) \\ \vdots \\ f(t_n) \end{bmatrix} \quad \text{and} \quad \alpha = \begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_5 \end{bmatrix}$$

From the handout, we obtain

$$\hat{\alpha} = (A^T A)^{-1} A^T b = \begin{bmatrix} -0.037 \\ 2.216 \\ 2.89 \\ -11.23 \\ 7.7 \\ -1.54 \end{bmatrix}$$

Inset plot here

(b) We write  $\hat{f}(t_i) = \alpha_1 \sin(\pi t_i) + \dots + \alpha_5 \sin(5\pi t_i)$

and form the matrices

$$A = \begin{bmatrix} \sin(\pi t_1) & \dots & \sin(5\pi t_1) \\ \vdots & & \vdots \\ \sin(\pi t_N) & \dots & \sin(5\pi t_N) \end{bmatrix}$$

$$b = \begin{bmatrix} f(t_1) \\ \vdots \\ f(t_N) \end{bmatrix} \quad \text{and} \quad \alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_5 \end{bmatrix}$$

From the handout, we obtain

$$\hat{\alpha} = (A^T A)^{-1} A^T b = \begin{bmatrix} 0.8106 \\ 0.00 \\ -0.0901 \\ 0.00 \\ 0.325 \end{bmatrix}$$

Insert plot here

Here is the MATLAB code

Insert Prob8.txt here



9. There is no "correct" choice for a basis. I chose  $\{1, t, t^2, t^3\}$ .

Repeating the results of Prob 8, I obtained

$$\hat{\alpha} = \begin{bmatrix} 0.0495 \\ -0.0847 \\ -1.8287 \\ 2.890 \end{bmatrix}$$

That is

$$\hat{f}(t) = 0.0495 - 0.0847t - 1.8287t^2 + 2.890t^3$$

I obtained

$$\left. \frac{d\hat{f}(t)}{dt} \right|_{t=0.3} \approx -0.4$$

Insert  
Figure here

Here is the MATLAB code

[Insert Prob 9.txt].